

P. Sri Sai Charan

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TECHNICAL SUMMARY

AIML undergraduate building full-stack AI systems that connect ML models to working backends and user-facing interfaces. Core focus: anomaly detection pipelines, explainability layers, REST/WebSocket APIs, and React dashboards. Approaches engineering with architecture-first thinking — structuring systems for clarity, traceability, and defensible design decisions.

EDUCATION

Vardhaman College of Engineering

B.Tech in Artificial Intelligence & Machine Learning

CGPA: 8.45 / 10

Hyderabad, India

Aug 2024 – May 2028

TECHNICAL SKILLS

Languages	Python, Java, C, JavaScript, SQL
AI / ML	Scikit-learn, TensorFlow, Pandas, NumPy, OpenCV, MediaPipe, Isolation Forest, Random Forest, Explainable AI (feature attribution)
Backend	Node.js, FastAPI, REST APIs, WebSockets, Prisma ORM, SQLite
Frontend	React, TypeScript, TailwindCSS, HTML, CSS
Tools	Git, GitHub, VS Code, Jupyter

PROJECTS

Sports Medicine & Anti-Doping Intelligence Platform

Jan 2025 – Present

React · TypeScript · TailwindCSS · Node.js · FastAPI · WebSockets · Prisma ORM · SQLite · Scikit-learn · Isolation Forest · Random Forest · OCR

- Architected a full-stack biomarker monitoring system: Node.js/FastAPI REST layer handles data ingestion and athlete profile management; ML models run server-side and return structured results consumed by the React frontend.
- Trained an Isolation Forest model on historical biomarker sequences to flag readings that deviate from each athlete's own rolling baseline; a Random Forest classifier then assigns a categorical risk level (low / medium / high) to each flagged record.
- Added an explainability layer using Scikit-learn's feature importances to surface which biomarker values drove each risk flag — outputs are rendered inline on the dashboard so analysts see reasoning, not just a score.
- Implemented WebSocket channels so the dashboard receives new alerts and biomarker updates without polling; alert state is managed in React context and persisted via Prisma ORM to SQLite.
- Built an OCR ingestion pipeline using pytesseract to extract structured values from uploaded lab report PDFs, reducing manual entry and normalizing data before model inference.

AI Forest Fire & Smoke Detection

2024

Python · OpenCV · Scikit-learn · Image Processing

- Built a CV classification pipeline for smoke and fire detection: applied HSV masking and morphological filtering to isolate candidate regions, then trained an ML classifier on extracted features; optimized threshold to reduce false negatives.

AI Athlete Performance Analyzer

2025

Python · Scikit-learn · Pandas · NumPy

- Constructed a feature set from raw training-session logs (load, volume, recovery metrics) and applied regression models to predict near-term performance trends; validated against a held-out split.

Shortify — URL Shortening Service

2024

Node.js · REST APIs · JavaScript

- Designed and implemented a backend service for slug generation, collision handling, and redirect resolution using a layered routing architecture.